



Special Factoring

Some polynomials follow specific patterns. It is a good idea to memorize these patterns to make factoring easier.

When you subtract two perfect squares, it is called a **difference of squares**.

$$a^2 - b^2 = (a + b)(a - b)$$

Example 1

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Factor: $x^2 - 49$
The square root of x^2 is x .
The square root of 49 is 7.
 $x^2 - 49 = (x + 7)(x - 7)$

Trinomial squares have two identical factors. These trinomials are perfect squares.

$$x^2 + 2ax + a^2 = (x + a)(x + a) \text{ or } (x + a)^2$$

$$x^2 - 2ax + a^2 = (x - a)(x - a) \text{ or } (x - a)^2$$

Example 2

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Factor: $x^2 + 6x + 9$
The square root of 9 is 3, and $3(2) = 6$. This fits the pattern of a trinomial square.
 $x^2 + 6x + 9 = (x + 3)(x + 3) \text{ or } (x + 3)^2$

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Lesson 2: Factoring

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